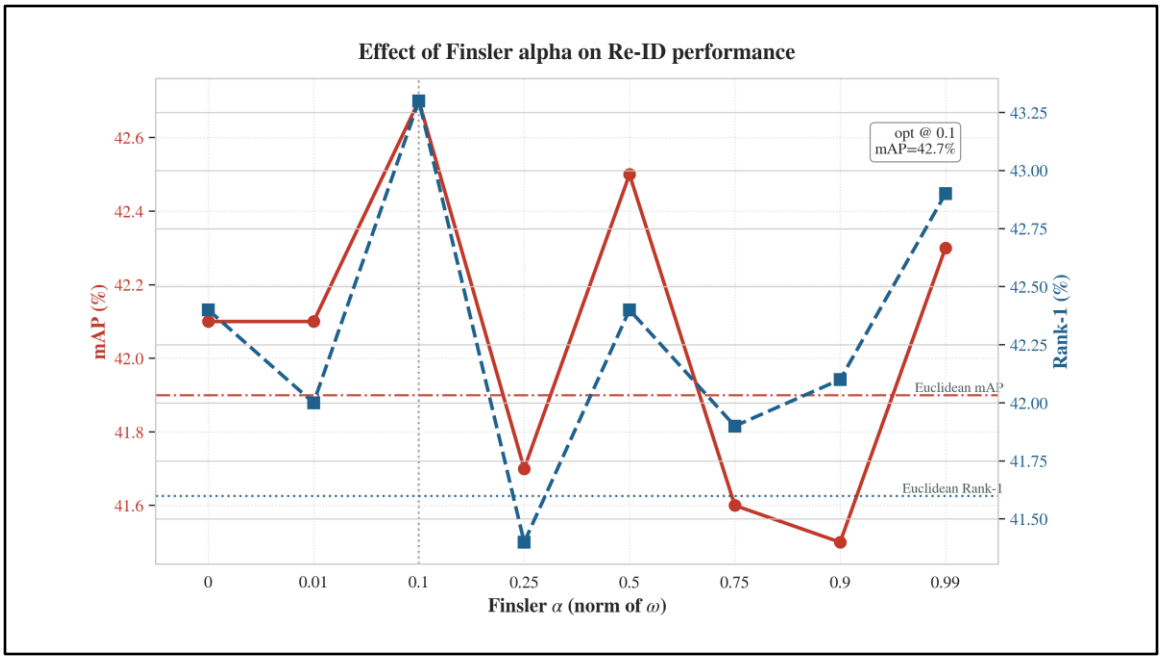
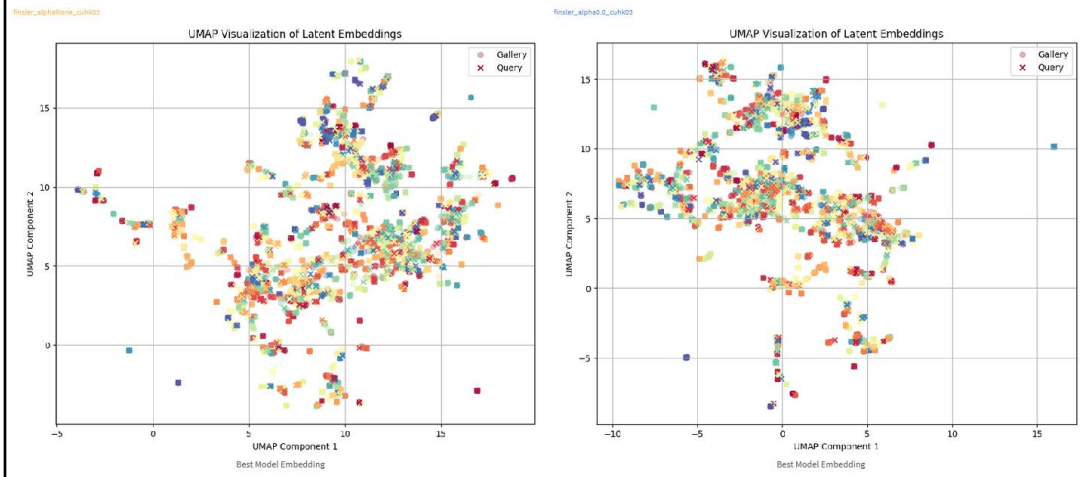


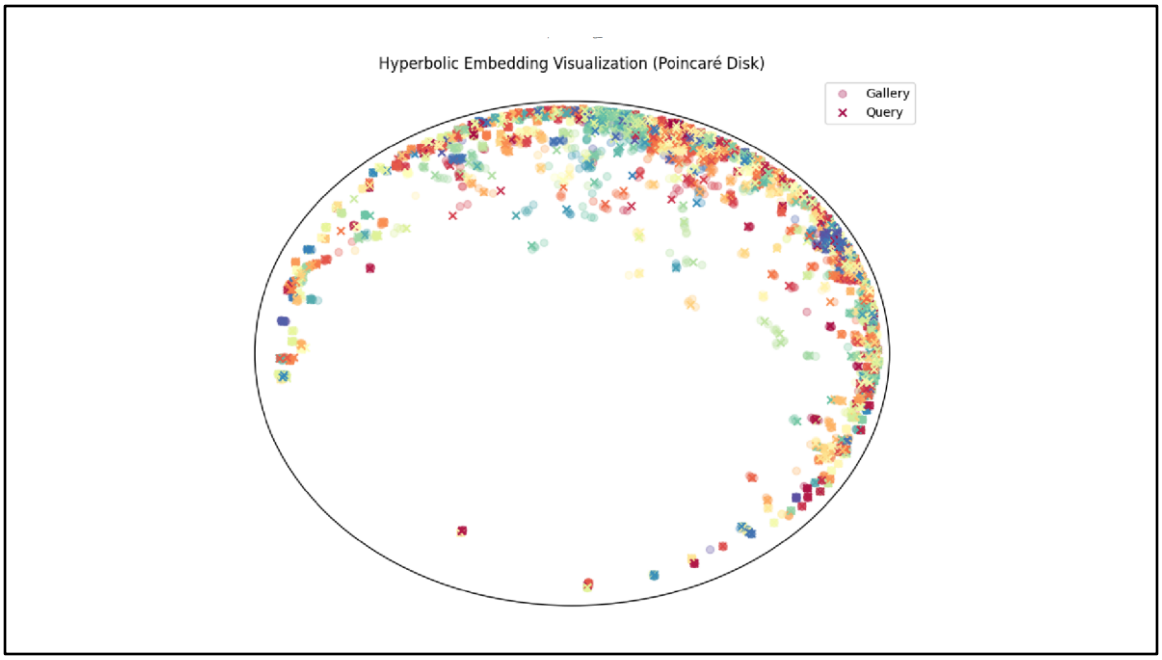
- Alpha 0.0 and alpha=None have different compute graphs, which can change behavior in kernel fusion and rounding under autocast
- This creates tiny numeric differences that accumulate over time
- Non-deterministic behavior from cudnn.benchmark=True

- Do five runs per configuration and average it out (smoothen out to make general configurations from it)
- Ablate the losses, maybe finsler affects this component most
- Maybe other components the model we can finslerize

```
# Base hyper-parameters (shared across jobs)
ARCH=resnet50            # Default
BATCH_SIZE=256           # Default
LAM=1.5                  # Default
KNN=10                   # Default
ITERS=500                # Default
WARMUP_STEP=10           # Default
EPOCHS=60                # Default
```



- Seed UMAP + Remove multiple processes that uses non-deterministic (some parallel things, e.g., number of jobs)
- Seed before UMAP and seed in UMAP itself
- Maybe use TSNE and seed everything
- Use ISOMAP (less likely to form cluster)
- Play with the hyperparameters
- Subsample the clusters think more about what I want to show



- Maybe PCA preprocessing
- See how imagenet embeds things
- There should be some visual clusters
- But here there are no clusters whatsoever
- 2000 embeddings should be okay

ID	Loss Function Configuration	Justification / What we want to show
1	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig}$	**Sanity Check**: Values should match between Euclidean and Hyperbolic variants*
2	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{triplet}}^{orig}$	Isolates the impact of the triplet loss + check training stability
3	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{align}}$	Isolates the impact of alignment loss (See Eq. 5) + check training stability
4	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{uniform}}$	Isolates the impact of uniformity loss (See Eq. 6) + check training stability
5	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{domain}}$	Isolates the impact of domain loss (See Eq. 7) + check training stability
6	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig}$	**Baseline w/o augmentations**: Isolates the effect of hyperbolic geometry on standard person Re-ID Training
7	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{ce}}^{aug}$	Cross-entropy on augmented images using same labels as originals
8	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{align}}$	Alignment between original and augmented hyperbolic embeddings (See Eq. 5)
9	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{uniform}}$	Uniformity of hyperbolic embeddings on the Poincaré ball (See Eq. 6)
10	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{domain}}$	Domain-specific uniformity with prototypes defined and updated in the Poincaré ball (See Eq. 8)
11	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{uniform}}$	Synergy between $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ as regularizers; impact of $\mathcal{L}_{\text{domain}}$
12	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{domain}}$	Synergy between $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{domain}}$ as regularizers; Impact of $\mathcal{L}_{\text{uniform}}$
13	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{domain}} + \mathcal{L}_{\text{uniform}}$	Synergy between $\mathcal{L}_{\text{domain}}$ and $\mathcal{L}_{\text{uniform}}$ as regularizers; impact of $\mathcal{L}_{\text{align}}$
14	$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}}^{orig} + \mathcal{L}_{\text{triplet}}^{orig} + \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{uniform}} + \mathcal{L}_{\text{domain}}$	Synergy between $\mathcal{L}_{\text{align}}$, $\mathcal{L}_{\text{uniform}}$ and $\mathcal{L}_{\text{domain}}$ as regularization terms on augmented images

- Do the first five and see how it is
- If triplet loss curve looks good, go to 5-10
- Do the ablation from BAU on Finsler
- Reduce the range of the hyperparameters of alpha
    - 0,, (0.01), 0.1, (0.3), 0.5
    - If 0.1, 0.5 are fluctuating, we stop there
    - Above 0.5 does not give good results
    - Ri

    - Right now its difficult to see how Finsler synergizes with BAU loss terms
    - Learnable alpha? – add hyperparameter here
    - One number output with sigmoid activation to constrain it from 0 to 1.